

Customer Journey Map

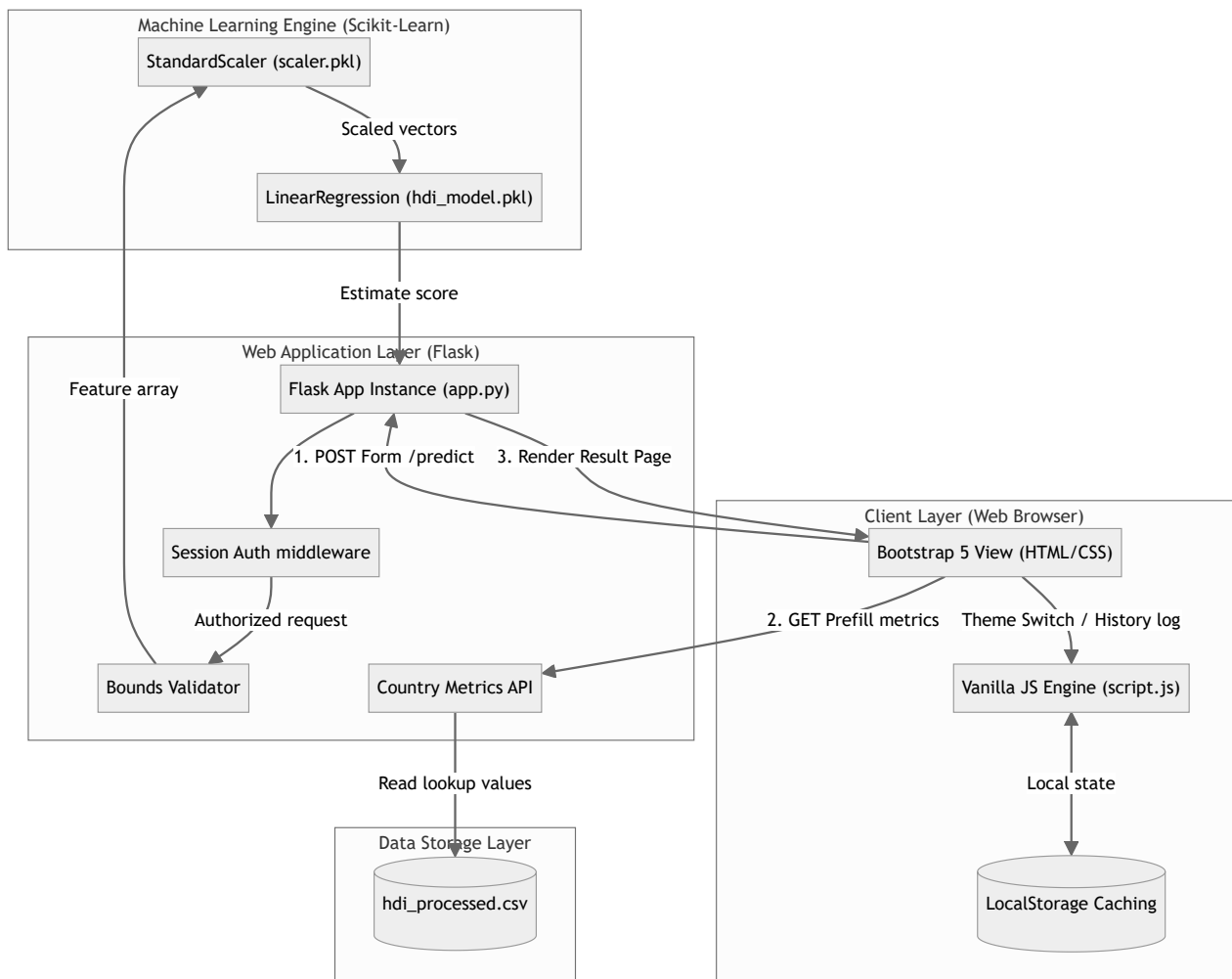
3. PROJECT DESIGN PHASE

Date	4 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well Being
Maximum Marks	4 Marks

Solution Architecture Specification

This document details the software architecture layers and interaction protocols behind our HDI Predictor system.

Solution Layers Diagram



Architecture Component Descriptions

- 1. Authentication Session Guard:** Intercepts requests using `app.before_request`. Validates if `session['logged_in']` is active. Redirects to `/login` if unauthenticated.
- 2. Bounds Validator:** Checks incoming query features (Life Expectancy, Schooling, GNI) against predefined ranges. Rejects out-of-bounds metrics with HTTP 400.
- 3. ML Pickles Pipeline:**
- 4. scaler.pkl:** Restores mean and variance offsets from the training dataset.
- 5. hdi_model.pkl:** Houses the weights matrix, multiplying standardized vectors to predict final index scores.
- 6. Local History Database:** JavaScript-controlled LocalStorage caching. Encapsulates predictions as JSON dictionaries and appends them to a list, avoiding SQL database setups.